

Ahmed Adil

Senior Software Engineer

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PROFESSIONAL SUMMARY

Forward-thinking Software Engineer with 4 years of enterprise experience specializing in complex system architecture, machine learning integrations, and scalable full-stack development. Currently pursuing an MS in Artificial Intelligence, combining strong theoretical knowledge of predictive modeling and computer vision with hands-on expertise in delivering enterprise-grade SaaS platforms. Proven ability to architect robust data pipelines, build multi-tenant cloud systems, and develop intelligent, automated solutions.

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER

2024 – Present

Xavor Corporation, Lahore, Pakistan

- **Architected an end-to-end RAG pipeline on Databricks** using PySpark and Azure OpenAI, transforming unstructured documents into queryable semantic vector embeddings.
- **Engineered a self-correcting Adaptive RAG workflow** with LangGraph, utilizing an "LLM-as-a-Judge" to autonomously evaluate retrieval accuracy and route to fallback data.
- **Built autonomous AI agents** using LangChain and Model Context Protocol (MCP) to dynamically orchestrate Databricks compute endpoints and execute vector searches without hardcoded logic.
- **Deployed a stateful ReAct agent** with persistent memory checkpointing, enabling the LLM to retain historical context and synthesize complex insights across multi-turn conversations.
- **Bridged the gap between complex backend architectures and scalable data pipelines**, ensuring enterprise systems are optimized for heavy computational workloads and future AI integrations.
- **Collaborated on AI based healthcare project with University of California**, Irvine and deployed a solution.
- Delivered and maintained a highly scalable, multi-tenant enterprise-grade system integrated with Kentico CMS, successfully driving the launch and onboarding of 20 distinct tenants within a two-year span.
- **Architected and co-developed a robust Digital Asset Management (DAM) system** on AWS utilizing serverless architecture to ensure secure, high-availability media handling for enterprise clients.
- Improved performance of a backend API system by migrating a .net based Legacy codebase to modern tech stack using AI assisted development.

SOFTWARE ENGINEER

2022 – 2024

Xavor Corporation, Lahore, Pakistan

- Scaled application infrastructure by integrating Next.js with Headless CMS capabilities to drastically enhance performance and content delivery.
- Devised and deployed a dynamic, tenant-based authentication flow, seamlessly and selectively implemented across specific tenants within a complex multi-tenant architecture.
- Engineered a custom CSS library from the ground up, significantly accelerating development speed and optimizing frontend workflows for the engineering team.
- Engineered and deployed a centralized repository of 300+ internal tools and utilities, drastically optimizing development workflows and cross-team operational efficiency.

EDUCATION

MASTERS IN ARTIFICIAL INTELLIGENCE
LUMS

2025 – Present

- **Core Focus:** Advanced machine learning architecture, big data processing, predictive modeling, agentic AI, and LLM application development using the LangChain ecosystem.
- **Key Coursework & Applied Concepts:** Neural Networks, Deep Learning, Support Vector Machines (SVM), Principal Component Analysis (PCA), K-means Clustering, Logistic Regression, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), prompt engineering, AI agents, and workflow orchestration.
- **Data Engineering Focus:** Gained hands-on experience in building and optimizing big data pipelines for large-scale data transformation, analytics, and LLM-powered data workflows.
- **Technologies Used:** Apache Spark, Delta Lake, Python (Data Science Stack), LangChain, LangGraph, LangSmith, vector databases, and modern LLM APIs.

BS-EE COMPUTER ENGINEERING
FAST-NU

2018 – 2022

CORE SKILLS

Artificial Intelligence & Machine Learning: Python, Neural Networks, Support Vector Machines (SVM), Principal Component Analysis (PCA), K-means Clustering, Logistic Regression, Computer Vision, Prompt Engineering, LangChain, LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), AI Agents, LLM Application Development, Model Context Protocol (MCP)

Data Engineering & Agentic AI: Apache Spark, Delta Lake, Data Pipeline Architecture, LLM Data Workflows, Vector Databases

Robotics & Autonomous Systems: ROS 2, Mapping Algorithms, Pathfinding

Software Engineering: JavaScript, TypeScript, React, Next.js, Electron.js, Node.js